



Fig. 4: Power LEDs and USB charging circuit

as the charging voltage to safely and efficiently charge the battery, thus maximising battery capacity and life.

Here, IC BQ24450 is configured as a simple constant-voltage float charge controller for the 6V, 4.5Ah battery. It is so flexible that it can be programmed in various configurations to suit different charging currents and voltages for batteries of different capacities. Only an external pass transistor and very few external passive components are required along with the IC to implement a charger for sealed lead-acid batteries.

Here, a BD240 pnp transistor (T1) is used to drive the battery charging power. Before proceeding to battery charger programming, please refer to the datasheet of BQ24450. Note that charging rate can be 0.05C to 0.3C. We will use charging rate ($I_{max-chg}$) = 0.13C = 600mA (approx).

In programming, there are parameters to implement a dual-level charger for the given battery chemistry. The first step is to decide the value of current in the voltage divider resistor string in float mode. It should be substantially higher than the input bias current in CE and VFB pins and the leakage current in STAT1 pin of BQ24450, but low enough so that the voltage on PGOOD pin does not introduce errors. A value of 50µA is suitable. Readers interested in calculations may go through the calculations_charging.doc file included in the EFX DVD of this month.

LED1 in the BQ24450 circuitry displays the battery charging status. Schottky diode D6 prevents the back current leakage from the battery. Thereafter the circuit feeds the lead-acid battery. Next, the battery rail goes

to the low-voltage detection circuit and IC UCC383 (IC4) in series with a mains mechanical load switch (S1).

The low-voltage-battery detection circuit is built around comparator LMP7300 (IC3), which has an internal reference of 2.048 volts. The voltage at which the battery should disconnect from the load is around 1.75V per cell, which means 5.25V for three-cell batteries.

The comparator has open-collector output, which goes low when the battery voltage drops below 5.25V. The comparator output goes to CT pin of IC4 in the next stage. Thus whenever the battery voltage goes below 5.25 volts, the comparator output goes low. This, in turn, switches IC4 off, shutting down all the loads connected to the battery and thus preventing the battery from discharging further.

The potential divider network comprising resistors R2 through R6 establishes 2.048V at the non-inverting terminal of comparator IC3 when the battery rail reaches 5.25V. It is compared with the voltage at the inverting terminal of the comparator, which is directly connected to the reference output of LMP7300. Further, a combined negative and positive hysteresis of about 300mV in total is programmed to LMP7300 through R7 and R8, so that the comparator output does not fluctuate when the battery voltage reaches the exact crossing point of the comparator's reference potential.

The UCC383 LDO is capable of driving loads up to 3A with a maximum dropout of 0.45 volts, which is excellent for this application. V_{IN} pin accepts the battery rail, which nominally varies in the range of 6 to 7.5 volts while charging and 5.25 to 6

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